

CLINISCOPE

Healthcare Record Contradiction Detector
Find the conflict. Trace the evidence. Keep humans in control.

TEAM [Add official team name]

Team leader: [Add team leader]

Team members: [Add member names]

Problem Statement

PS-11R3: Healthcare Record Contradiction Detector. The framing below is the team's interpretation.

01 THE PROBLEM

Patient information is spread across many documents. Allergies, doses, diagnoses and lab results can differ, and spotting that by hand is slow.

02 TARGET USERS

Intended users: clinicians reviewing records, medical-records (HIM) teams, clinical quality and patient-safety reviewers, and authorized staff reconciling records.

03 CURRENT CHALLENGES

- Fragmented: statements sit in separate files
- Context: dates, negation, dose changes
- Traceability: each conflict needs its source

04 THE GAP / IMPACT

A warning is not enough. Reviewers need the original evidence, its context (dates, history) and a controlled way to record their decision.

ONE-LINE PROBLEM: How can reviewers find and investigate conflicts across medical records with clear source evidence and human oversight?

Proposed Solution

An evidence-first review prototype: documents in, traceable findings out, decisions made by people.

SOLUTION OVERVIEW

CLINISCOPE extracts text from PDF, TXT and DOCX records, finds allergy, medication, diagnosis and lab statements with rules, compares them across a case and flags potential conflicts with verified source quotes for human review.

UNIQUE VALUE / USP

Every finding is traceable to an exact source quote. Dated or documented changes are separated from true conflicts, and people make the final call.

01 DOCUMENT INGESTION

PDF (text layer), TXT and DOCX are validated and extracted in the browser.

02 EVIDENCE DETECTION

Rule-based comparison flags potential conflicts with source quotes.

03 REVIEW & AUDIT

Decisions need a reason; every action is kept in an audit log.

EXPECTED BENEFIT: Conflicts become easier to find and investigate, with evidence and human judgment kept.

Technical Approach & Architecture

Everything runs in the reviewer's browser: no backend server, no AI/LLM and no OCR in the current build.



SYSTEM COMPONENTS

Extraction (pdf.js, mammoth) → statement rules → contradiction detection → evidence check → review workflow, stored with Dexie.

DATA FLOW & VALIDATION

Uploads are type- and size-checked. A finding is kept only if its quotes match the source text; dates can reclassify it.

ARCHITECTURE NOTE: Extraction, detection and review are separate, unit-tested modules.

Technology Stack

Technologies verified in package.json and the CI workflow, and the role each one plays.

FRONTEND

React 18, TypeScript, Vite and Tailwind CSS for the review UI.

BACKEND

None: a browser-side TypeScript service layer runs the workflow.

DATABASE

IndexedDB via Dexie: cases, files, findings and audit events.

AI / CORE LOGIC

No AI/LLM: deterministic TypeScript rules for detection.

INTEGRATIONS

No external APIs. pdfjs-dist (PDF) and mammoth (DOCX) run locally.

DEPLOYMENT & TOOLS

Git, GitHub Actions (Vitest, Playwright) and GitHub Pages.

WHY THIS STACK? A light, testable static web app: records are processed and stored in the browser, so no server is needed. OCR, AI or a backend would be added only when required.

Feasibility & Viability

A working, tested prototype today, with a staged and cautious path toward real-world use.

TECHNICAL FEASIBILITY

- Required tools and skills: React, TypeScript, rule design, pdf.js, mammoth, Vitest, Playwright.
- Data / infrastructure: synthetic records and browser storage only; no real patient data.
- Prototype status: deployed; 34 unit and 8 end-to-end tests pass. No OCR or login yet.
- Risk + mitigation: false positives → verified quotes, uncertainty labels, human review.

BUSINESS / OPERATIONAL VIABILITY

- Target users and value: authorized reviewers who need evidence, not just alerts.
- Adoption plan: synthetic prototype → test set → clinician feedback → approved pilot.
- Cost and maintenance: hosting, processing, future storage or AI APIs, testing.
- Revenue (future only): institutional licensing or integration. No partners today.

KEY TAKEAWAY: A working prototype with a practical, validation-first path to evidence-based review.

Sustainability, Scalability & Impact

Scalability items are planned, not built. Impact is a goal to be measured in evaluation.

LONG-TERM SUSTAINABILITY

Modular engine, regression tests in CI, versioned schema, and unsupported files flagged openly rather than guessed.

SCALABILITY

Planned: OCR for scans, more formats, evidence-checked AI help, secure backend with login and multi-user review.

EXPECTED IMPACT

Aims to reduce manual searching, make investigations traceable and record review decisions consistently.

FUTURE ROADMAP



01 Prototype



02 Pilot testing



03 Improve



04 Deploy & scale

SUCCESS METRIC: Precision/recall on labelled records, % findings with evidence, reviewer agreement.

Conclusion

CLINISCOPE helps reviewers find potential conflicts across healthcare records, trace each one to source evidence, and record decisions with human oversight.

Find the conflict. Trace the evidence. Keep humans in control.

THANK YOU

[Add team email] • github.com/anirudhleetcodes-max/cliniscope-healthcare-contradiction-detector
Live demo: anirudhleetcodes-max.github.io/cliniscope-healthcare-contradiction-detector (scan QR)

